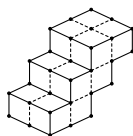


HK IMO Training 2018-2019
Proofs in Combinatorics
15 December 2018

I Miscellaneous Problems on Proofs in Combinatorics

1. In a chess tournament, every pair of players played a match and there is no draw. A directed cycle of length n is n people that can be seated on a round table such that everyone won against the person immediately clockwise to him. Prove that if there is a directed cycle of any length, there is a directed cycle of length 3.
2. (ARO 17) In country some cities are connected by oneway flights(There are no more then one flight between two cities). City A called *available* for city B , if there is flight from B to A , maybe with some transfers. It is known, that for every 2 cities P and Q exist city R , such that P and Q are available from R . Prove, there exists a city A , such that every city is available from A .
3. (Tournament of Towns) 100 Queens are placed on a 100×100 chessboard so that no two attack each other. Prove that each of four 50×50 corners of the board contains at least one Queen.
4. (Tournament of Towns 02) An infinite sequence of natural number $\{x_n\}_{n \geq 1}$ is such that x_{n+1} is obtained by adding one of the non-zero digits of x_n to itself. Show this sequence contains an even number.
5. (IMO 05 P2, SL N2) Let $a_1, a_2, \dots$ be a sequence of integers with infinitely many positive and negative terms. Suppose that for every positive integer n the numbers $a_1, a_2, \dots, a_n$ leave n different remainders upon division by n . Prove that every integer occurs exactly once in the sequence $a_1, a_2, \dots$.
6. (SL07 C2) A rectangle D is partitioned in several (≥ 2) rectangles with sides parallel to those of D . Given that any line parallel to one of the sides of D , and having common points with the interior of D , also has common interior points with the interior of at least one rectangle of the partition; prove that there is at least one rectangle of the partition having no common points with D 's boundary.
7. A staircase-brick with 3 steps of width 2 is made of 12 unit cubes. Determine all integers n for which it is possible to build a cube of side n using such bricks.



8. (SL04 C1) There are 10001 students at an university. Some students join together to form several clubs (a student may belong to different clubs). Some clubs join together to form several societies (a club may belong to different societies). There are a total of k societies. Suppose that the following conditions hold:
 - (i) Each pair of students are in exactly one club.
 - (ii) For each student and each society, the student is in exactly one club of the society.
 - (iii) Each club has an odd number of students. In addition, a club with $2m + 1$ students is in exactly m societies.
Find all possible values of k .
9. (Putnam 2013 A2) Let S be the set of all positive integers that are not perfect squares. For $n \in S$, consider choices of integers $a_1, a_2, \dots, a_r$ such that $n < a_1 < a_2 < \dots < a_r$ and $n \cdot a_1 \cdot a_2 \cdots a_r$ is a perfect square, and let $f(n)$ be the minimum of a_r over all such choices. Show that f is injective. i.e., $f(a) \neq f(b)$ whenever $a \neq b$. (For example, $2 \cdot 3 \cdot 6$ is a perfect square, while $2 \cdot 3 \cdot 2 \cdot 4$, $2 \cdot 5 \cdot 2 \cdot 3 \cdot 4$, $2 \cdot 3 \cdot 5 \cdot 2 \cdot 4 \cdot 5$, and $2 \cdot 3 \cdot 4 \cdot 5$ are not, and so $f(2) = 6$.)
10. (Russian Math Olympiad 88) Ivan has a 52-card deck. He draws the cards from the deck one by one, without putting them back in the deck. Every time before drawing a card he guesses the suit of the card he will draw. He decides to always guess the suit that occurs most frequently in the remaining deck (if there are several such suits, he chooses any one of them). Prove that he will guess the right suit at least 13 times.
11. (SL15 C1) In Lineland there are $n \geq 1$ towns, arranged along a road running from left to right. Each town has a left bulldozer (put to the left of the town and facing left) and a right bulldozer (put to the right of the town and facing right). The sizes of the $2n$ bulldozers are distinct. Every time when a left and right bulldozer confront each other, the larger bulldozer pushes the smaller one off the road. On the other hand, bulldozers are quite unprotected at their rears; so, if a bulldozer reaches the rear-end of another one, the first one pushes the second one off the road, regardless of their sizes. Let A and B be two towns, with B to the right of A . We say that town A can sweep town B away if the right bulldozer of A can move over to B pushing off all bulldozers it meets. Similarly town B can sweep town A away if the left bulldozer of B can move over to A pushing off all bulldozers of all towns on its way. Prove that there is exactly one town that cannot be swept away by any other one.
12. (IMO 00 P4, SL C1) A magician has one hundred cards numbered 1 to 100. He puts them into three boxes, a red one, a white one and a blue one, so that each box contains at least one card. A member of the audience selects two of the three boxes, chooses one card from each and announces the sum of the numbers on the chosen cards. Given this sum, the magician identifies the box from which no card has been chosen. How many ways are there to put all the cards into the boxes so that the trick always works? (Two ways are considered different if at least one card is put into a different box.)
13. (SL13 N5) Fix an integer $k \geq 2$. Two players, called Ana and Banana, play the following game of numbers. Initially, some integer $n \geq k$ gets written on the blackboard. Then they take moves in turn, with Ana beginning. A player making a move erases the number m just written on the blackboard and replaces it by some number m' with $k \leq m' < m$ that is coprime to m . The first player who cannot move anymore loses. An integer $n \geq k$ is called *good* if Banana has a winning strategy when the initial number is n , and *bad* otherwise. Consider two integers $n, n' \geq k$ with the property that each prime number $p \leq k$ divides n if and only if it divides n' . Prove that either both n and n' are good or both are bad.

14. (IMO 14 P5, SL N3) For each positive integer n , the Bank of Cape Town issues coins of denomination $\frac{1}{n}$. Given a finite collection of such coins (of not necessarily different denominations) with total value at most $99 + \frac{1}{2}$, prove that it is possible to split this collection into 100 or fewer groups, such that each group has total value at most 1.

II Extra problems

Below are short interesting problems (just in case you get bored!) that have nothing to do with the main theme. They all have short solutions.

1. A maze is an 8×8 board with some adjacent squares separated by walls, so that any two squares can be connected by a path not meeting any wall. Given a command LEFT, RIGHT, UP, DOWN, a pawn makes a step in the corresponding direction unless it encounters a wall or an edge of the chessboard. Faust writes a program consisting of a finite sequence of commands and gives it to Mephistopheles, who then constructs a maze and places the pawn on one of the squares. Can Faust write a program which guarantees the pawn will visit every square despite Mephistopheles's efforts?
2. (Russia 61) A number is written in each of the squares of an $m \times n$ grid. It is allowed to switch the sign of all numbers from one row or one column. Prove that eventually it is possible to get a grid, in which the sums of the numbers in every column and every row are non-negative.
3. On a square board divided into 15×15 little squares there are 15 rooks that do not attack each other. Then each rook makes one move like that of a knight. Prove that after this is done a pair of rooks will necessarily attack each other.
4. **** (ARO 11) A 2010×2010 board is divided into corner-shaped figures of three cells. Prove that it is possible to mark one cell in each figure such that each row and each column will have the same number of marked cells.
5. ***** (ARO) In a country of n cities, an express train runs both ways between any two cities. For any train, ticket prices either direction are equal, but for any different routes these prices are different. Prove that the traveler can select the starting city, leave it and go on, successively, $n - 1$ trains, such that each fare is smaller than that of the previous fare. (A traveler can enter the same city several times.)

* means a difficult problem and ** means a very difficult problem.

Appendix

Problems are arranged (roughly) ascending order of difficulty.

Most of the problems demand no special skills, only a capacity to visualize and organize things clearly. These problems never require constructions.

Of course, you will still need the following core ideas, essentially to all of mathematics (except geometry):

- induction,
- divide and conquer,
- contradiction by infinite descent, and
- proof by contradiction

Apart from the methods above, you need as well the following:

- to work on a concrete small case example(s) and
- to visualise the existing problem, best through a diagram.

Bibliography

Arvin Leung's collection:

- ◊ Strong Combinatorics Collection
- ◊ IMO shortlist compilations

(Both are available on the official Google Drive).

Russian-Style Problems (Alexander Remorov, Canada IMO training)